

Kevin Wang

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EDUCATION

University of California, Santa Cruz - B.S. Electrical Engineering | Expected 2027 | GPA: 3.6

Relevant Coursework: Analog Electronics, High-Speed Digital Design, Energy Conversion & Control, Signals & Systems, Logic Design, Electronic Design Automation

EXPERIENCE

Formula Slug (UCSC FSAE) - Low Voltage Electrical Shutdown Engineer

Sep 2024 - Jun 2025

- Worked on the design and **testing** of shutdown circuitry for the vehicle's tractive system.
- Tested proposed circuit designs in **LTspice** to check logic, timing, and behavior under fault conditions.
- Analyzed **BMS** and **IMD** fault logic, relay control, and supporting analog circuits using **schematics** and **datasheets**.
- Used simulation and circuit analysis to compare component choices and help guide final design decisions.
- Documented circuit operation, simulation results, design decisions, and overall subsystem behavior.
- Worked with electrical and mechanical team members to make sure shutdown requirements were properly integrated into the vehicle.

Digsur Tech - Outreach & Communications Intern

Sep 2025 - Dec 2025

- Presented drone-based surveying and 3D rendering services to prospective clients.
- Explained drone capabilities, applications, and operating use cases during customer presentations.
- Gained experience with commercial drone systems used for surveying, geospatial data collection, and 3D environment reconstruction.

Engineering & Physics Tutor

Oct 2025 - Present

- Tutored circuit labs involving **hardware** wiring, schematic reading, measurements, and **debugging**.
- Taught the behavior and use of op-amps, diodes, potentiometers, DACs, flip-flops, and inverter ICs.
- Helped students use multimeters, **oscilloscopes**, **spectrum analyzers**, and **AC/DC power supplies** during lab work.
- Guided students through signal tracing, voltage measurements, waveform analysis, and physical circuit troubleshooting.
- Tutored signals and systems, differential equations, electrostatics, and classical mechanics.

PROJECTS

High-Power DC Electronic Load Test Bench - In Development

- Developing a portable electronic load for DC power-source **testing** with a target range up to **150 W**, **30 V**, and **10 A**.
- Designing the **PCB** in **KiCad** around three parallel power **MOSFETs**, **current-sense** circuitry, op-amp **feedback**, protection, and low-voltage support electronics.
- Planning forced-air heatsinking, enclosure-mounted controls, and a compact assembly for repeatable bench and portable **testing**.
- Designing an adaptable connector interface for **USB-C**, **USB**, barrel-jack, and other DC source connections.

Programmable Electronic Load and Test Platform

- Designed and built a programmable constant-current load for **testing** DC power sources at different load currents.
- Used an LM324 op-amp, MOSFET, sense resistor, and DAC reference to create closed-loop current control.
- Connected an **ESP32** microcontroller, **MCP4725 DAC**, OLED display, and rotary encoder for current adjustment and system **feedback**.
- Wrote embedded C firmware for **I2C** communication, DAC control, display output, and rotary encoder input.
- Traced **I2C** and **hardware** problems through voltage measurements, continuity checks, clock-rate changes, wiring checks, and grounding tests.
- Found a missing ground trace on the DAC module caused by a manufacturing defect.

High-Speed Digital Circuit Demonstration PCB

- Designed and assembled a 2-layer **PCB** to demonstrate ground bounce, channel loss, NEXT, and FEXT.
- Used return paths, parasitic inductance, coupling, and switching behavior to guide the circuit and **PCB** design.
- Created test structures with different trace spacing, routing lengths, and geometries to examine individual signal-integrity effects.
- Used an **oscilloscope** and **spectrum analyzer** to measure waveform distortion and frequency-domain behavior.
- Added SMA connectors and dedicated test points to make high-frequency measurements easier to repeat.
- Checked the schematic, **PCB** layout, design rules, and manufacturing files before fabrication and **testing**.

Audio Amplifier PCB

- Built and tested a fabricated two-layer analog audio amplifier **PCB**.
- Turned **schematics**, **datasheets**, and component requirements into a manufacturable board using OrCAD and Allegro.
- Created custom footprints from package dimensions, pad geometry, drill sizes, and component spacing.
- Routed analog signal and power connections with attention to grounding, current paths, and noise-sensitive circuitry.
- Checked the schematic, **PCB** layout, footprints, and Gerber files before fabrication.

Basys3 VGA Game

- Built a simplified Subway Surfers-style game on a Basys 3 FPGA using behavioral Verilog and state-machine logic.
- Created 60 FPS VGA output with pixel-level graphics for player movement, tracks, and multi-color objects.
- Generated randomized train obstacles across three tracks with different lengths, positions, and spawn intervals.
- Added pixel-based collision detection, jump timing, reset control, and an optional invincibility mode.
- Used FPGA buttons and switches for movement, jumping, reset, and other gameplay controls.

SKILLS

PCB / EDA: OrCAD Capture, Allegro **PCB** Designer, PSpice/LTspice, Altium Designer, **KiCad**, GerbTool

Programming / Simulation: Verilog, Python, C, MATLAB, Simulink, ANSYS Maxwell, dSPACE

Hardware / Testing: Oscilloscopes, Spectrum Analyzers, Function Generators, Multimeters, **AC/DC Power Supplies**, Breadboarding, Signal Tracing

Electrical Engineering: Analog Electronics, Op-Amp & MOSFET Circuits, **Feedback** & Current Regulation, Digital Logic & State Machines, Signals & Systems, **Signal Integrity**, High-Speed Digital Design, Power & Energy Conversion, **I2C** Interfaces, **PCB** Design, Circuit Simulation, **Hardware**

Bring-Up & Debugging